

# 1ST PROOF PROBLEM

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## 1. THE PROBLEM

Let  $m$  and  $n$  be positive integers and let  $R_n^{(m)}$  be the coinvariant algebra with  $m$  sets of  $n$  variables.

The Hilbert series of an  $m$ -multigraded algebra  $A$  is

$$\text{Hilb}(A; q_1, \dots, q_m) = \sum_{i_1, \dots, i_m \geq 0} \dim(A_{(i_1, \dots, i_m)}) q_1^{i_1} \cdots q_m^{i_m},$$

where  $A_{(i_1, \dots, i_m)}$  denotes a multigraded component of the algebra, with grading variables  $q_1, \dots, q_m$ .

Let  $s_\lambda(q_1, \dots, q_m)$  denote a Schur polynomial. For  $R_n^{(m)}$ , it is known that its Hilbert series may be expressed by

$$\text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda} c_\lambda(n) s_\lambda(q_1, \dots, q_m),$$

for some nonnegative universal series coefficients  $c_\lambda(n)$  which do not depend on  $m$ .

For any  $n$ , give a combinatorial interpretation of the coefficients  $c_\lambda(n)$  when  $\lambda$  is a hook shape partition.

## 2. THE SOLUTION

**2.1. Background.** The **coinvariant ring with  $m$  set of variables** is defined by

$$(1) \quad R_n^{(m)} = \mathbb{C}[x_1^{(1)}, \dots, x_n^{(1)}, \dots, x_1^{(m)}, \dots, x_n^{(m)}] / \langle \mathbb{C}[x_1^{(1)}, \dots, x_n^{(1)}, \dots, x_1^{(m)}, \dots, x_n^{(m)}]_{+}^{\mathfrak{S}_n} \rangle,$$

where  $\mathbb{C}[x_1^{(1)}, \dots, x_n^{(1)}, \dots, x_1^{(m)}, \dots, x_n^{(m)}]_{+}^{\mathfrak{S}_n}$  denotes the polynomials without constant term, which are invariant under the diagonal action of  $\mathfrak{S}_n$  permuting each set of variables.

The multigraded Hilbert series of  $R_n^{(m)}$  is defined by

$$(2) \quad \text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) := \sum_{r_1, \dots, r_m \geq 0} \dim \left( (R_n^{(m)})_{r_1, \dots, r_m} \right) q_1^{r_1} \cdots q_m^{r_m}.$$

Following from the work of Bergeron [1], this Hilbert series may be written in the form

$$(3) \quad \text{Hilb}(R_n^{(m)}; q_1, \dots, q_m) = \sum_{\lambda, \ell(\lambda) \leq m} c_\lambda(n) s_\lambda(q_1, \dots, q_m),$$

for some nonnegative integer coefficients  $c_\lambda(n)$ , which are independent of  $m$ . It is important to note that while they are independent of  $m$ ,  $s_\lambda(q_1, \dots, q_m)$  can evaluate to 0 if  $\lambda$  has too many parts relative to  $m$ . Given a fixed  $n \geq 1$ , all of the possible  $c_\lambda(n)$  coefficients that could possibly appear do appear once  $m \geq n$ , by [1]. Thus we can reduce to the case where  $m = n$ :

$$(4) \quad \text{Hilb}(R_n^{(n)}; q_1, \dots, q_n) = \sum_{\lambda, \ell(\lambda) \leq n} c_\lambda(n) s_\lambda(q_1, \dots, q_n).$$

Now in the problem, we are asked to give an explicit combinatorial formula for the  $c_\lambda(n)$  when  $\lambda$  is a hook shape partition. A partition  $\lambda$  which satisfies  $\lambda_2 \leq 1$  is called a **hook shape**. By diagonal supersymmetry (see Proposition 2), conjectured by [1] and proven by [3], we are able to convert this

to a problem about the superspace coinvariant ring. This is because all of the hook shapes  $\lambda$  which satisfy  $\ell(\lambda) \leq n$  can be determined from this setting.

The **superspace coinvariant ring** is defined by

$$(5) \quad R_n^{(1,1)} = \mathbb{C}[x_1, \dots, x_n, \theta_1, \dots, \theta_n] / \langle \mathbb{C}[x_1, \dots, x_n, \theta_1, \dots, \theta_n]_{+}^{\mathfrak{S}_n} \rangle,$$

where  $\mathbb{C}[x_1, \dots, x_n, \theta_1, \dots, \theta_n]_{+}^{\mathfrak{S}_n}$  denotes the polynomials without constant term, which are invariant under the diagonal action of  $\mathfrak{S}_n$  permuting each set of variables. The  $x_i$  are commutative variables, and the  $\theta_i$  are anticommutative variables. Commuting variables commute with all variables, while anticommuting variables anticommute with all anticommuting variables, that is,  $\theta_i \theta_j = -\theta_j \theta_i$ .

The superspace coinvariant ring  $R_n^{(1,1)}$  is bigraded, where each  $x_i$  contributes to the  $q$ -degree and each  $\theta_i$  contributes to the  $u$ -degree. Let  $(R_n^{(1,1)})_{r,s}$  denote the subspace of  $R_n^{(1,1)}$  which has  $q$ -degree  $r$  and  $u$ -degree  $s$ . Then the bigraded Hilbert series is defined by

$$(6) \quad \text{Hilb}(R_n^{(1,1)}; q; u) := \sum_{r,s \geq 0} \dim \left( (R_n^{(1,1)})_{r,s} \right) q^r u^s.$$

Let  $[k]_q$  denote the  $q$ -integer  $1 + q + \dots + q^{k-1}$ . Define the  $q$ -factorial number  $[k]_q!$  by  $[k]_q! = [k]_q [k-1]_q \cdots [2]_q [1]_q$ , and let  $[0]_q! := 1$ . Then define the  $q$ -binomial coefficients by

$$(7) \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]_q!}{[k]_q! [n-k]_q!}.$$

Ordered set partitions of  $\{1, \dots, n\}$  with  $k$  blocks are sequences of nonempty sets  $w = (S_1/S_2/\dots/S_k)$  such that the disjoint union of the sets  $S_i$  is  $\{1, \dots, n\}$ . We denote the set of all ordered set partitions of  $\{1, \dots, n\}$  into  $k$  blocks by  $\text{OSP}(n, k)$ . Following Sagan–Swanson [5, Section 5.1], define the inversion set on ordered set partitions by

$$(8) \quad \text{Inv}(w) = \{(s, S_j) \mid s \in S_i \text{ for some } i < j \text{ and } s \geq \min S_j\},$$

and the inversion statistic by  $\text{inv}(w) = \# \text{Inv}(w)$ .

Recall the  **$q$ -Stirling numbers of the second kind**  $S[n, k]$ , which are defined by the initial conditions  $S[0, k] = \delta_{0,k}$  and  $S[n, 0] = \delta_{n,0}$ , along with the recurrence relation

$$(9) \quad S[n, k] = S[n-1, k-1] + [k]_q S[n-1, k]$$

for  $0 \leq k \leq n$ .

Rhoades and Wilson proved the following explicit formula for the bigraded Hilbert series of  $R_n^{(1,1)}$ .

**Theorem 1** ([4]). *For any  $n \geq 1$ ,*

$$(10) \quad \text{Hilb}(R_n^{(1,1)}; q; u) = \sum_{k=0}^n [k]_q! S[n, k] u^{n-k}.$$

Denote a Schur function by  $s_\lambda(q_1, \dots, q_k)$  and a skew Schur function by  $s_{\lambda/\mu}(q_1, \dots, q_k)$ . A **super Schur function**  $s_\lambda(q_1, \dots, q_k/u_1, \dots, u_j)$  is defined by

$$(11) \quad s_\lambda(q_1, \dots, q_k/u_1, \dots, u_j) = \sum_{\nu \subseteq \lambda} s_\nu(q_1, \dots, q_k) s_{\lambda' \setminus \nu'}(u_1, \dots, u_j),$$

where  $\lambda'$  denotes the transpose of the partition  $\lambda$  (see for example [2, Sections A.2.2 and 3.2]).

We recall diagonal supersymmetry for Hilbert series of coinvariant rings, specialized to the superspace coinvariant ring. Let  $P(k, j, n)$  denote the set of all partitions which satisfy  $\lambda_{k+1} \leq j$  and the length  $\ell(\lambda) \leq n$ .

**Proposition 2** ([3]). *Fix a positive integer  $n$ . For partitions  $\lambda$  with  $\ell(\lambda) \leq n$ , there exist nonnegative integer coefficients  $c_\lambda(n)$  such that the bigraded Hilbert series of  $R_n^{(1,1)}$  is*

$$(12) \quad \text{Hilb}(R_n^{(1,1)}; q; u) = \sum_{\lambda \in P(1,1,n)} c_\lambda(n) s_\lambda(q/u).$$

Note that  $P(1,1,n)$  consists of all hook shape partitions of length at most  $n$ . This is a key insight in the simplification of the problem.

The super Schur function has a simple formula for hook shape partitions.

**Lemma 3.** *If  $\lambda$  is  $(a, 1^b)$  for some  $a \geq 1$  and  $b \geq 0$ , then  $s_\lambda(q/u) = q^a u^b + q^{a-1} u^{b+1}$ . If  $\lambda$  is  $\emptyset$ , then  $s_\lambda(q/u) = 1$ . If  $\lambda$  is not contained in a hook shape, then  $s_\lambda(q/u) = 0$ .*

We recall the following.

**Proposition 4** ([5, Theorem 5.1]). *For  $n, k \geq 0$ , we have*

$$(13) \quad [k]_q! S[n, k] = \sum_{w \in \text{OSP}(n, k)} q^{\text{inv}(w)}.$$

In other words,

$$(14) \quad \langle q^i \rangle [k]_q! S[n, k] = \#\{w \in \text{OSP}(n, k) \mid \text{inv}(w) = i\}.$$

Now we recall the involution  $\phi$  of Sagan–Swanson [5, Section 5.1]. Given an ordered set partition  $w = (S_1/S_2/\dots/S_k)$ , the ordered set partition  $w$  is called **splittable** at  $M = \max S_i$ , for a fixed  $i \in \{1, \dots, k\}$ , if  $|S_i| > 1$ . The ordered set partition  $w$  is called **mergeable** at  $M = \max S_i$ , for a fixed  $i \in \{1, \dots, k-1\}$ , if  $|S_i| = 1$  and  $M > \max S_{i+1}$ .

Define a function  $\phi$  on ordered set partitions of  $[n]$ . Given  $w = (S_1/S_2/\dots/S_k)$ , find the largest  $M$  such that  $w$  is either splittable or mergeable at  $M$ , if such an  $M$  exists. We say that

- $w$  is type I if  $w$  is mergeable at  $M$ ;
- $w$  is type II if  $w$  is splittable at  $M$ ;
- $w$  is type III if no  $M$  exists.

Then

$$(15) \quad \phi(w) = \begin{cases} (S_1/S_2/\dots/S_{i-1}/S_i \cup S_{i+1}/S_{i+2}/\dots/S_k) & \text{if } w \text{ is type I,} \\ (S_1/S_2/\dots/S_{i-1}/\{M\}/S_i - \{M\}/S_{i+1}/\dots/S_k) & \text{if } w \text{ is type II,} \\ w & \text{if } w \text{ is type III.} \end{cases}$$

It follows that  $\phi$  is an involution. As merging removes one inversion and one block and splitting adds one inversion and one block then if  $w$  has  $\text{blocks}(w)$  blocks and  $\text{inv}(w)$  inversions, then  $n - \text{blocks}(w) - \text{inv}(w)$  is preserved by  $\phi$ . Finally  $\phi$  is a sign-reversing involution on all ordered set partitions (which are not fixed points) with the definition  $\text{sgn}(w) = (-1)^{n - \text{blocks}(w)}$ .

**2.2. Proofs.** In this subsection, we prove the desired combinatorial formula. First, we express the universal series coefficients  $c_\lambda(n)$  that we need in terms of a signed enumeration of certain ordered set partitions.

**Lemma 5.** *For  $a \geq 1$  and  $b \geq 0$ , the coefficients  $c_{(a, 1^b)}(n)$  from equation (12) may be computed by*

$$(16) \quad c_{(a, 1^b)}(n) = \sum_{i=0}^b (-1)^i \#\{w \in \text{OSP}(n, n-b+i) \mid \text{inv}(w) = a+i\}.$$

*Proof.* By equations (12) and (10), we have that

$$(17) \quad \sum_{\lambda \in P(1,1,n)} c_\lambda(n) s_\lambda(q/u) = \sum_{d=0}^n [d]_q! S[n, d] u^{n-d}.$$

Then by Lemma 3, we have

$$(18) \quad 1 + \sum_{\substack{a \geq 1, \\ b \geq 0}} c_{(a,1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}) = \sum_{d=0}^n [d]_q! S[n, d] u^{n-d}.$$

By equation (14),

$$(19) \quad 1 + \sum_{\substack{a \geq 1, \\ b \geq 0}} c_{(a,1^b)}(n) (q^a u^b + q^{a-1} u^{b+1}) = \sum_{d=0}^n \sum_{i \geq 0} q^i \# \{w \in \text{OSP}(n, d) \mid \text{inv}(w) = i\} u^{n-d}.$$

By extracting the coefficients of  $q^a u^b$  from both sides, we find that, for  $b \geq 1$ ,

$$(20) \quad c_{(a,1^b)}(n) + c_{(a+1,1^{b-1})}(n) = \# \{w \in \text{OSP}(n, n-b) \mid \text{inv}(w) = a\}$$

and for  $b = 0$ ,

$$(21) \quad c_{(a)}(n) = \# \{w \in \text{OSP}(n, n) \mid \text{inv}(w) = a\}.$$

Finally, we solve for  $c_{(a,1^b)}(n)$  to establish equation (16). □

We desire to improve upon Lemma 5 to get a manifestly positive formula for the coefficients  $c_{(a,1^b)}(n)$ .

For  $0 \leq b \leq \lfloor \frac{\ell-1}{2} \rfloor$ , define the set

$$(22) \quad S_{n,\ell,b} = \bigcup_{c=0}^b \{w \in \text{OSP}(n, n-c) \mid \text{inv}(w) = \ell - b - c\}.$$

This set is a reindexing of the union of the sets that appears in equation (16).

While  $\phi$  is an involution on the set of all ordered set partitions, it does not necessarily fix  $S_{n,\ell,b}$ . We describe when this happens in the following lemma.

**Lemma 6.** *If  $w \in S_{n,\ell,b}$ , then:*

- (i) *if  $w$  is of type I,  $w \in \text{OSP}(n, n-c)$  and  $\text{inv}(w) = \ell - 2c$ , then  $\phi(w) \notin S_{n,\ell,b}$ ;*
- (ii) *if  $w$  is otherwise of type I, then  $\phi(w) \in S_{n,\ell,b}$ ;*
- (iii) *if  $w$  is of type II, then  $\phi(w) \in S_{n,\ell,b}$ ;*
- (iv)  *$w$  cannot be of type III.*

*Proof.* Let  $b$  be fixed. The condition on  $\text{inv}(w)$  for  $S_{n,\ell,b}$  requires that  $\ell - 2b \leq \text{inv}(w) \leq \ell - b$ , by considering the values  $c$  can take on.

- (i) If  $w$  is of type I,  $w \in \text{OSP}(n, n-c)$  and  $\text{inv}(w) = \ell - 2c$ , this means that  $c = b$ , so  $\text{inv}(w) = \ell - 2b$ . As  $\phi$  applied to an ordered set partition of type I lowers the number of inversions by 1,  $\text{inv}(\phi(w)) = \ell - 2b - 1$ , so  $\phi(w) \notin S_{n,\ell,b}$ .
- (ii) If  $w$  is otherwise of type I, then  $c < b$ . We get that  $\text{inv}(\phi(w)) = \ell - b - c - 1 \geq \ell - 2b$ . Thus  $\phi(w) \in S_{n,\ell,b}$ .
- (iii) If  $w$  is of type II, then  $\text{blocks}(w) < n$ , so  $c > 0$ , which implies that  $\text{inv}(w) < \ell - b$ . Thus  $\text{inv}(\phi(w)) \leq \ell - b$  and  $\phi(w) \in S_{n,\ell,b}$ .
- (iv) Suppose towards a contradiction that  $w \in S_{n,\ell,b}$  is of type III, then  $c = 0$  and  $\ell = b$ . Since  $\ell \geq 1$ , then  $b > \lfloor \frac{\ell-1}{2} \rfloor$ , contradicting that  $b \leq \lfloor \frac{\ell-1}{2} \rfloor$ .

□

Let us collect those ordered set partitions in  $S_{n,\ell,b}$  which are not paired under  $\phi$  as

$$(23) \quad T_{n,\ell,b} = \{w \in \text{OSP}(n, n-b) \mid w \text{ of type I and } \text{inv}(w) = \ell - 2b\}.$$

Now we are ready to state the answer to the problem.

**Theorem 7.** *For  $a \geq 1$  and  $b \geq 0$ , the coefficients  $c_{(a,1^b)}(n)$  from equation (12) may be computed in a manifestly positive manner by*

$$(24) \quad c_{(a,1^b)}(n) = \#\{w \in \text{OSP}(n, n-b) \mid \text{inv}(w) = a, w \text{ is type I}\}.$$

*Proof.* Use the  $\phi$  involution of Sagan–Swanson on

$$(25) \quad S_{n,a+2b,b} = \bigcup_{i=0}^b \{w \in \text{OSP}(n, n-b+i) \mid \text{inv}(w) = a+i\}.$$

Those points in  $S_{n,a+2b,b}$  which are not paired under  $\phi$  are

$$(26) \quad T_{n,a+2b,b} = \{w \in \text{OSP}(n, n-b) \mid \text{inv}(w) = a, w \text{ is type I}\}.$$

□

**2.3. Example.** In this subsection, we give an example of our solution at  $n = 4$ .

| Coefficient      | Value | OSPs                                                      |
|------------------|-------|-----------------------------------------------------------|
| $c_{(1)}(4)$     | 3     | 2 1 3 4   1 3 2 4   1 2 4 3                               |
| $c_{(1,1)}(4)$   | 3     | 1 4 23   12 4 3   3 12 4                                  |
| $c_{(1,1,1)}(4)$ | 1     | 4 123                                                     |
| $c_{(2)}(4)$     | 5     | 2 1 4 3   1 3 4 2   2 3 1 4   3 1 2 4   1 4 2 3           |
| $c_{(2,1)}(4)$   | 5     | 4 1 23   13 4 2   2 4 13   3 4 12   4 12 3                |
| $c_{(3)}(4)$     | 6     | 2 3 4 1   3 2 1 4   2 4 1 3   3 1 4 2   4 1 2 3   1 4 3 2 |
| $c_{(3,1)}(4)$   | 4     | 4 13 2   23 4 1   4 2 13   4 3 12                         |
| $c_{(4)}(4)$     | 5     | 4 1 3 2   4 2 1 3   3 4 1 2   3 2 4 1   2 4 3 1           |
| $c_{(4,1)}(4)$   | 1     | 4 23 1                                                    |
| $c_{(5)}(4)$     | 3     | 4 2 3 1   4 3 1 2   3 4 2 1                               |
| $c_{(6)}(4)$     | 1     | 4 3 2 1                                                   |

FIGURE 1. A table showing the coefficients  $c_\lambda(n)$  and their contributing ordered set partitions (OSPs) given by Theorem 7. Aside from  $c_\emptyset(4) = 1$ , for all other  $\lambda$  not covered in the table, those coefficients are  $c_\lambda(4) = 0$ .

One can compute that

$$\begin{aligned}
\text{Hilb}(R_n^{(1,1)}; q; u) &= (q^6 + 3q^5 + 5q^4 + 6q^3 + 5q^2 + 3q + 1) + (q^5 + 4q^4 + 9q^3 + 11q^2 + 8q + 3)u \\
&\quad + (q^3 + 4q^2 + 6q + 3)u^2 + u^3 \\
&= 1 + c_{(1)}(4)s_{(1)}(q/u) + c_{(2)}(4)s_{(2)}(q/u) + c_{(3)}(4)s_{(3)}(q/u) + c_{(4)}(4)s_{(4)}(q/u) \\
&\quad + c_{(5)}(4)s_{(5)}(q/u) + c_{(6)}(4)s_{(6)}(q/u) + c_{(1,1)}(4)s_{(1,1)}(q/u) + c_{(2,1)}(4)s_{(2,1)}(q/u) \\
&\quad + c_{(3,1)}(4)s_{(3,1)}(q/u) + c_{(4,1)}(4)s_{(4,1)}(q/u) + c_{(1,1,1)}(4)s_{(1,1,1)}(q/u).
\end{aligned}$$

## REFERENCES

- [1] François Bergeron, The bosonic-fermionic diagonal coinvariant modules conjecture, preprint (2020), <https://arxiv.org/abs/2005.00924>.
- [2] Shun-Jen Cheng and Weiqiang Wang, Dualities and representations of Lie superalgebras, Graduate Studies in Mathematics, vol. 144, American Mathematical Society, Providence, RI, 2012.
- [3] John Lentfer, Diagonal supersymmetry for coinvariant rings, preprint (2025), <https://arxiv.org/abs/2505.14885>.
- [4] Brendon Rhoades and Andrew Timothy Wilson, The Hilbert series of the superspace coinvariant ring, Forum Math. Pi 12 (2024), 35 (English), Id/No e16.
- [5] Bruce E. Sagan and Joshua P. Swanson,  $q$ -Stirling numbers in type  $B$ , Eur. J. Comb. 118 (2024), 35, Id/No 103899

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